

Integrating Mastercard and Visa Card Issuance into Entrust Infrastructure

Preparation and configuration guide for Entrust Key Manager, EMV Profile Manager, and Card Wizard

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1. Purpose and Scope

This document describes the procedure for preparing and configuring the Entrust software stack so that a bank can issue Mastercard or Visa card products. The work is primarily a key ceremony: the correct keys must be created in the correct token table with the correct attributes, and the correct certificates must be imported in the correct sequence.

The complete procedure is summarised below:

```
Create token tables -> Generate issuer key pair -> Build CA request (ZIP)
-> Bank submits request to Mastercard -> Issuer certificate (.CEF) returned
-> Import CA certificate (.hep + .sep) -> Import issuer certificate (.CEF)
-> Import request/session keys (2DES, two components)
-> Import EMV profile and PQDF tool files -> Ready for issuance
```

Approximately 50 per cent of the integration effort is complete once the CA certificate and the issuer certificate have been imported successfully. The remaining work consists of importing the operational keys and the EMV profile files.

2. Prerequisites

2.1 Entrust portals used

All three portals of the Entrust ecosystem are used in this procedure.

No.	Portal	Role in this procedure
1	Key Manager Portal	Token tables, key pairs, certification authority requests, certificate and key imports
2	EMV Profile Manager Portal	EMV kernel and profile configuration for the card applet
3	Card Wizard Portal	Card personalization workflow and PQDF tool-file generation

2.2 Conditions to confirm with the client before starting

- The client operates **separate token tables for testing (UAT) and production (live)**; refer to section 3.1.
- The bank's **BIN number** is available. The BIN is used as the key owner.
- A bank **HOST operator** is available to generate 2DES keys **with two key components**.
- Access to the **Mastercard portal** is available in order to download the .hep and .sep files.

3. Procedure

3.1 Create separate token tables

Some clients store all key material in a single token table. This is workable but not advisable, because test and production material are then held in the same container. Two separate token tables must be created: one for testing (UAT) and one for production (live).

Create a new token table from the Key Manager Portal:

```
Entrust Key Manager Portal
  Setup
    HSM Configurations
      Right-click the HSM -> Login
        Create a new token table -> Assign a name
```

The token table is created once it appears in the HSM tree.

3.2 Generate the issuer key pair (Issuer_PK / Issuer_SK)

This step creates the Issuer Public Key (Issuer_PK), which is used to build the certification authority request, and the matching Issuer Private Key (Issuer_SK).

Critical. This is the most error-prone step of the entire integration; approximately 80 per cent of certification authority requests fail because of incorrect parameters at this point. Verify every attribute before generating the keys.

Navigate as follows and apply the parameters in the table below:

```
Key Manager Portal
  Home
    Select the correct token table -> Login
      Log in as Security Officer (SO)
        Generate Key Pair (side bar) -> Apply the parameters below
```

Parameter	Value	Comment
Key type	Any RSA-based key type	Naming depends on the HSM key-type set
Key length	1976 bits	As specified for Mastercard in this procedure
Owner	The bank's BIN number	Binds the key material to the issuer identity
Version	06 = live, EF = UAT	The version field identifies the environment
Usages	Encrypt, Wrap, Verify	All three usages are required
Attributes	Extractable, Exportable, Deletable	All three attributes are required
Exponent	03	Standard RSA public exponent

Note. Issuer_PK and Issuer_SK are generated with an identical procedure and identical parameters; only the key name differs.

3.3 Generate and submit the certification authority request

Generate the certification authority request that will be submitted to Mastercard:

```
Key Manager Portal
  Certificates
    Request a Certificate
      Mastercard
        Select the Issuer Public Key created in section 3.2
          Generate Request Certificate -> (ZIP file)
```

The request is transferred as follows:

1. Entrust generates the request as a ZIP archive.
2. The ZIP archive is handed to the bank, which submits it to Mastercard.
3. Mastercard returns the issuer certificate as a .cEF file.

Important. The .cEF file on its own is not sufficient to go live with Mastercard. Two further files must also be obtained; both are downloaded from the Mastercard portal. This is a common misunderstanding at banks.

Artifact	Format	Source	Purpose
Issuer certificate	.cEF	Mastercard reply to the CA request	Imported as the issuer certificate (section 3.4)
Hardware encryption program	.hep	Download from the Mastercard portal	Used to build the CA certificate in Entrust (section 3.4)
Security program	.sep	Download from the Mastercard portal	Used to build the CA certificate in Entrust (section 3.4)

All three files are required: .cEF, .hep, and .sep.

3.4 Import the CA certificate and the issuer certificate

Once all three files have been received, import them into Entrust in the following order.

Step 1 — CA certificate (HEP and SEP)

```
Key Manager Portal
  Import Certificate
    CA Certificate
      Select both the .hep and the .sep file -> Import
```

Step 2 — Issuer certificate

```
Key Manager Portal
  Import Certificate
    Issuer Certificate
      Select the .CEF file -> Import
```

Note. Once both imports have completed successfully, approximately 50 per cent of the integration is finished.

3.5 Import the required keys into the token table

The next stage is to request and import the operational keys (request and session keys) into the token table. Entrust reports which keys are required; this list is passed to the bank, which generates the keys with its HOST.

Critical. Instruct the bank to generate 2DES keys with two components. Entrust requires at least two key components in order to import a key.

When the key components have been received, import them as follows:

```
Key Manager Portal
  Import Secret Key or Key Pair
    Import Key with Components
      Enter the plain key components
        Enter the components a second time -> Generate Key
```

3.6 Import the EMV profile and PQDF tool files

Once the keys are in place, Entrust provides the following deliverables:

Deliverable	Purpose
EMV profile file	Configuration of the card applet and kernel
PQDF tool files	Personalization data tool files used for card issuance

Both are imported by following the guidelines supplied with the files.

Note. Minor adjustments to the configuration files are normally required to match the customer's card design and applet requirements before the first production run.

4. Verification Checklist

The integration may be signed off only when every item below has been confirmed.

- [] Two token tables exist: one for UAT and one for production.
- [] Issuer_PK and Issuer_SK were generated with the correct BIN owner, version (06 / EF), usages, attributes, and exponent 03.
- [] The CA request ZIP file was generated and handed to the bank.
- [] The bank submitted the request and Mastercard returned the issuer certificate (.cef).
- [] The .hep and .sep files were downloaded from the Mastercard portal.
- [] The CA certificate was imported (both .hep and .sep).
- [] The issuer certificate was imported (.cef).
- [] The required keys were requested from the bank as 2DES keys with two components each.
- [] The keys were imported using Import Key with Components (components entered twice).
- [] The EMV profile was imported.
- [] The PQDF tool files were imported and tuned for the customer's cards.

5. Known Failure Modes and Mitigations

No.	Failure mode	Consequence	Mitigation
1	Incorrect key-pair parameters (usages, attributes, exponent, or version)	Approximately 80 per cent of CA request failures originate here	Re-verify the parameter table in section 3.2 before generating the keys
2	Only the .cef file is obtained	The integration stalls at personalization	Obtain all three files: .cef, .hep, and .sep
3	A single token table is shared by UAT and production	Operationally impractical; test material is exposed to production processes	Always create separate token tables (section 3.1)
4	The bank generates keys with a single component	Entrust cannot import the key	Instruct the bank to generate 2DES keys with two components
5	EMV or PQDF files are imported without tuning	Cards behave incorrectly for the specific customer design	Follow the vendor guidelines and tune the files per customer card

6. Revision History

Rev.	Date	Description	Author
1.0	Initial release	First issue of the implementation guide, prepared from field notes covering Entrust Key Manager, EMV Profile Manager, and Card Wizard.	MinThutaSawNaing

